OFF-ROAD SCENE SEMANTIC SEGMENTATION USING U-NET WITH RESNET34

Team Lead: Deekshita Malla

Team Member: Tippa Gayathri

ABSTRACT

This project focuses on semantic segmentation of off-road scenes using deep learning techniques. Specifically, we use a U-Net architecture with a ResNet34 encoder to classify each pixel into one of ten terrain categories. The model is trained using a combination of CrossEntropy Loss and Dice Loss to handle class imbalance.

Experimental Results show that the model performs well on dominant classes while maintaining reasonable generalization.

1. INTRODUCTION

Semantic segmentation is a critical task in computer vision where each pixel is classified into a category. It is widely used in autonomous driving, robotics, and agriculture.

Off-road environments are complex due to irregular shapes, textures, and lighting variations. This makes segmentation challenging compared to structured road scenes.

In this project, we use deep learning techniques to perform pixel-wise classification of off-road images.

2. DATASET DESCRIPTION

The dataset consists of RGB images and segmentation masks.

Each image is resized to 256x256 and normalized.

There are 10 classes including trees, grass, rocks, and sky.

The dataset is imbalanced, with certain classes like trees dominating most images.

Data augmentation techniques like flipping are applied to improve generalization.

3. METHODOLOGY

We use the U-Net architecture which consists of an encoder and decoder.

The encoder (ResNet34) extracts features while the decoder reconstructs segmentation maps.

Skip connections help preserve spatial information. These connections improve localization by retaining fine-grained spatial details.Pretrained weights improve convergence speed.

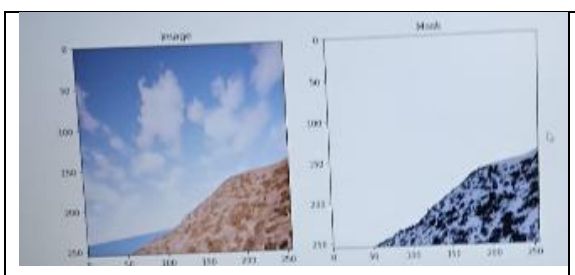
4. IMPLEMENTATION

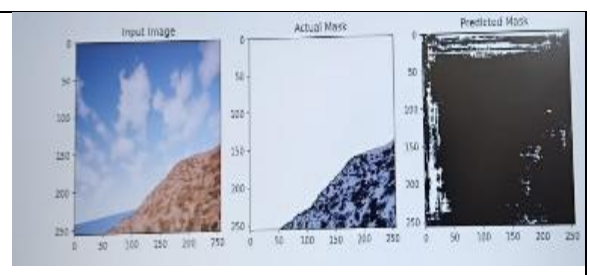
The model is implemented using PyTorch and segmentation_models_pytorch.

Training is performed using Adam optimizer with learning rate 0.0005.

The loss function is a combination of CrossEntropy and Dice Loss.

Initially, the model was trained for a few epochs to observe its performance.





Later, mask preprocessing was improved and training epochs were gradually increased along with better visualization techniques.

The following section presents the final improved version after training for 50 epochs. and We expect improved accuracy when training is performed on a GPU.

Due to storage limitations, the model was trained using VS Code instead of Google Colab.

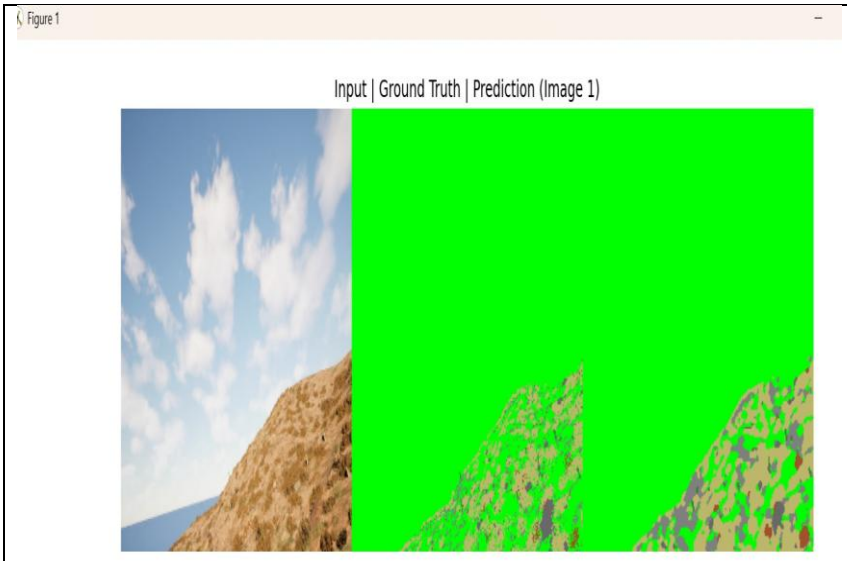
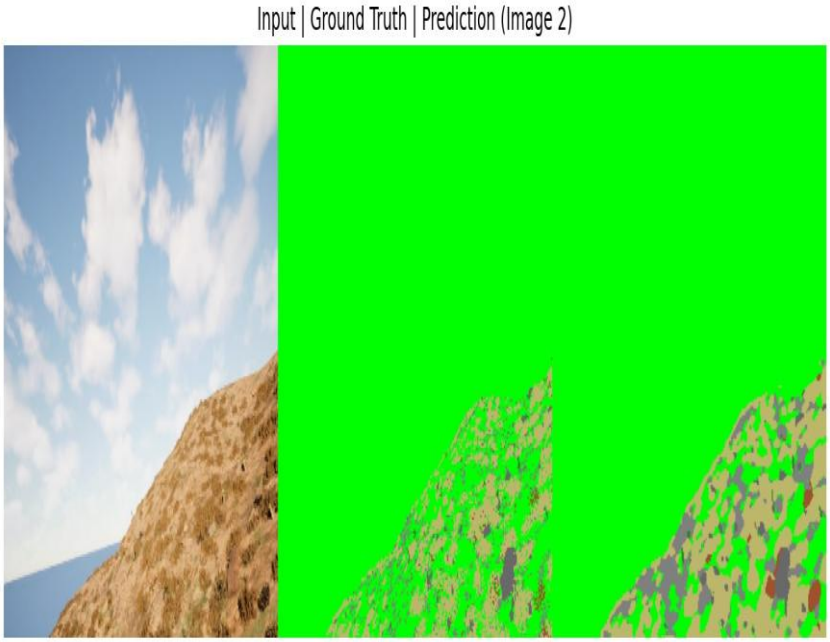
THE OUTPUT PROGRESS:

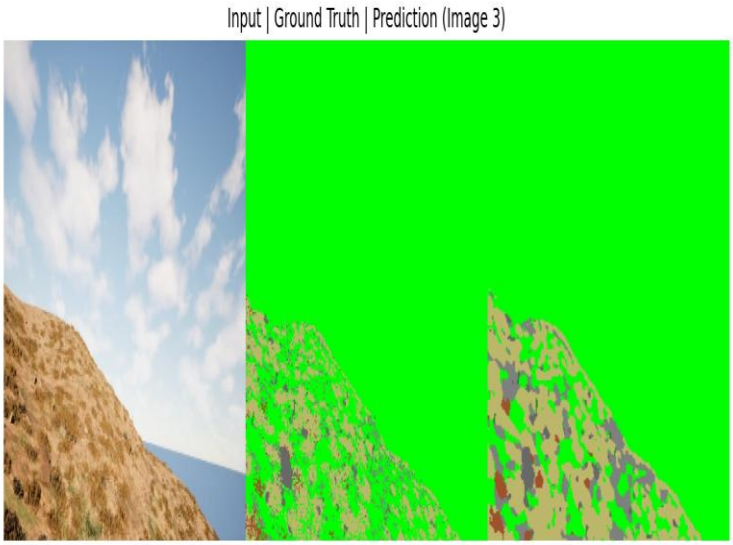
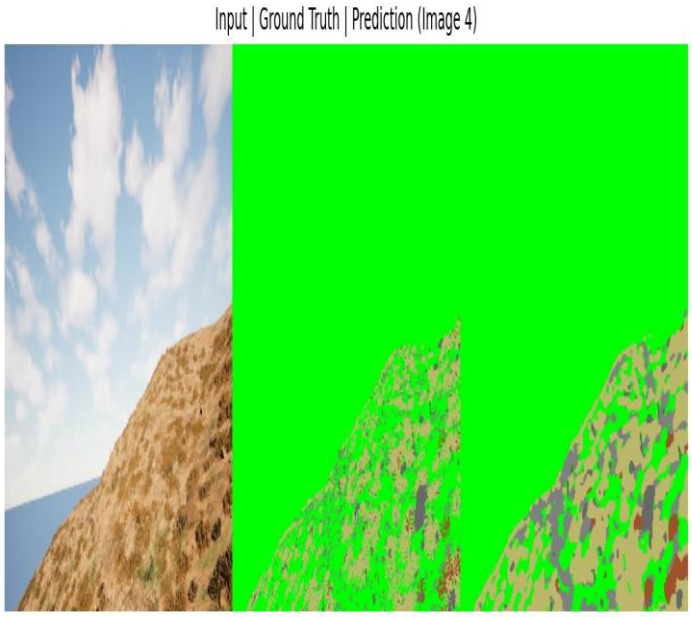
```
PS C:\Users\Deekshitha\Downloads\Offroad_Segmentation_Training_Dataset\Offroad_Segmentation_Training_Dataset> & "C:\Program Files\Python314\python
exe" c:/Users/Deekshitha/Downloads/Offroad_Segmentation_Training_Dataset/Offroad_Segmentation_Training_Dataset/SegmentationDataset.py
Epoch 1/50 | Loss: 2.7541 | Val Acc: 62.38% | IoU: 0.1362
Epoch 2/50 | Loss: 2.0551 | Val Acc: 65.04% | IoU: 0.2288
Epoch 3/50 | Loss: 1.8722 | Val Acc: 54.12% | IoU: 0.1981
Epoch 4/50 | Loss: 1.7214 | Val Acc: 62.34% | IoU: 0.2429
Epoch 5/50 | Loss: 1.6422 | Val Acc: 64.03% | IoU: 0.2648
Epoch 6/50 | Loss: 1.6040 | Val Acc: 63.72% | IoU: 0.2693
Epoch 7/50 | Loss: 1.5366 | Val Acc: 66.21% | IoU: 0.2924
Epoch 8/50 | Loss: 1.4921 | Val Acc: 68.97% | IoU: 0.2934
Epoch 9/50 | Loss: 1.4518 | Val Acc: 64.11% | IoU: 0.3039
Epoch 10/50 | Loss: 1.4222 | Val Acc: 67.13% | IoU: 0.3097
Epoch 11/50 | Loss: 1.4300 | Val Acc: 69.12% | IoU: 0.3201
Epoch 12/50 | Loss: 1.4184 | Val Acc: 65.34% | IoU: 0.2865
Epoch 13/50 | Loss: 1.3986 | Val Acc: 65.13% | IoU: 0.3055
Epoch 14/50 | Loss: 1.3582 | Val Acc: 65.58% | IoU: 0.2855
Epoch 15/50 | Loss: 1.3472 | Val Acc: 66.76% | IoU: 0.3185
Epoch 16/50 | Loss: 1.3443 | Val Acc: 65.35% | IoU: 0.3133
Epoch 17/50 | Loss: 1.3192 | Val Acc: 70.07% | IoU: 0.3616
Epoch 18/50 | Loss: 1.2953 | Val Acc: 67.88% | IoU: 0.3605
Epoch 19/50 | Loss: 1.2831 | Val Acc: 69.30% | IoU: 0.3863
Epoch 20/50 | Loss: 1.2577 | Val Acc: 69.14% | IoU: 0.3872
Epoch 21/50 | Loss: 1.2656 | Val Acc: 72.13% | IoU: 0.4026
Epoch 22/50 | Loss: 1.2466 | Val Acc: 69.75% | IoU: 0.3943
Epoch 23/50 | Loss: 1.2062 | Val Acc: 71.22% | IoU: 0.4067
Epoch 24/50 | Loss: 1.2090 | Val Acc: 67.21% | IoU: 0.3792
Epoch 25/50 | Loss: 1.2083 | Val Acc: 70.54% | IoU: 0.3998
Epoch 26/50 | Loss: 1.1963 | Val Acc: 71.72% | IoU: 0.4110
Epoch 27/50 | Loss: 1.1866 | Val Acc: 69.72% | IoU: 0.4034
Epoch 28/50 | Loss: 1.1938 | Val Acc: 69.62% | IoU: 0.3939
Epoch 29/50 | Loss: 1.1747 | Val Acc: 70.51% | IoU: 0.4004
Epoch 30/50 | Loss: 1.1786 | Val Acc: 70.42% | IoU: 0.4082
Epoch 31/50 | Loss: 1.1744 | Val Acc: 66.68% | IoU: 0.3804
```

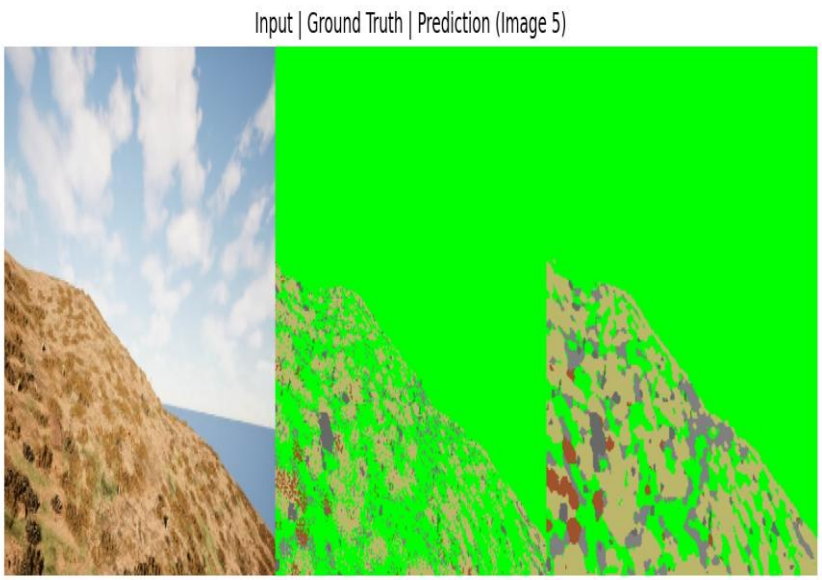
```
Epoch 32/50 | Loss: 1.1619 | Val Acc: 68.32% | IoU: 0.3740
Epoch 33/50 | Loss: 1.1639 | Val Acc: 70.83% | IoU: 0.4102
Epoch 34/50 | Loss: 1.1571 | Val Acc: 69.68% | IoU: 0.3968
Epoch 35/50 | Loss: 1.1458 | Val Acc: 69.83% | IoU: 0.3951
Epoch 36/50 | Loss: 1.1416 | Val Acc: 69.45% | IoU: 0.3909
Epoch 37/50 | Loss: 1.1346 | Val Acc: 69.23% | IoU: 0.3913
Epoch 38/50 | Loss: 1.1324 | Val Acc: 69.49% | IoU: 0.4058
Epoch 39/50 | Loss: 1.1322 | Val Acc: 71.78% | IoU: 0.4190
Epoch 40/50 | Loss: 1.1299 | Val Acc: 74.78% | IoU: 0.4550
Epoch 41/50 | Loss: 1.1217 | Val Acc: 72.31% | IoU: 0.4162
Epoch 42/50 | Loss: 1.1248 | Val Acc: 71.52% | IoU: 0.4200
Epoch 43/50 | Loss: 1.1216 | Val Acc: 70.84% | IoU: 0.4061
Epoch 44/50 | Loss: 1.1230 | Val Acc: 69.14% | IoU: 0.4071
Epoch 45/50 | Loss: 1.1200 | Val Acc: 69.71% | IoU: 0.4120
Epoch 46/50 | Loss: 1.1204 | Val Acc: 69.86% | IoU: 0.4169
Epoch 47/50 | Loss: 1.1127 | Val Acc: 67.80% | IoU: 0.3977
Epoch 48/50 | Loss: 1.1098 | Val Acc: 70.29% | IoU: 0.4164
Epoch 49/50 | Loss: 1.1052 | Val Acc: 70.12% | IoU: 0.4155
Epoch 50/50 | Loss: 1.1184 | Val Acc: 70.96% | IoU: 0.4076

Image 1 GT Percentages:
Trees: 90.55%
Lush Bushes: 0.07%
Dry Grass: 6.63%
Dry Bushes: 0.32%
Ground Clutter: 1.56%
Flowers: 0.00%
Logs: 0.00%
```

PREDICTION CLASS DISTRIBUTION AND ANALYSIS:

	Image1 Pred Percentages: Trees: 85.78% Lush Bushes: 0.07% Dry Grass: 8.94% Dry Bushes: 0.55% Ground Clutter: 3.84% Flowers: 0.00% Logs: 0.00% Rocks: 0.82% Landscape: 0.00% Sky: 0.00%
	Image 2 Pred Percentages: Trees: 83.87% Lush Bushes: 0.06% Dry Grass: 10.34% Dry Bushes: 0.54% Ground Clutter: 4.38% Flowers: 0.00% Logs: 0.00% Rocks: 0.80% Landscape: 0.00% Sky: 0.00%

	Image 3 Pred Percentages: Trees: 82.78% Lush Bushes: 0.08% Dry Grass: 12.41% Dry Bushes: 0.63% Ground Clutter: 3.29% Flowers: 0.00% Logs: 0.00% Rocks: 0.82% Landscape: 0.00% Sky: 0.00%
	Image 4 Pred Percentages: Trees: 80.03% Lush Bushes: 0.09% Dry Grass: 13.73% Dry Bushes: 0.99% Ground Clutter: 4.11% Flowers: 0.00% Logs: 0.00% Rocks: 1.04% Landscape: 0.00% Sky: 0.00%

	Image 5 Pred Percentages: Trees: 74.67% Lush Bushes: 0.08% Dry Grass: 17.40% Dry Bushes: 1.41% Ground Clutter: 5.27% Flowers: 0.00% Logs: 0.00% Rocks: 1.18% Landscape: 0.00% Sky: 0.00%

5. RESULTS AND ANALYSIS

The model was trained for 50 epochs. Loss decreased steadily and validation accuracy improved. The best validation accuracy reached approximately **74.78%**, and the model achieved a maximum Intersection over Union (IoU) score of approximately **0.45**, indicating good overlap between predicted and ground truth segmentation. This accuracy refers to pixel-wise classification performance. Model performance stabilizes after 40 epochs.

6. VISUAL RESULTS

Predicted segmentation masks closely resemble ground truth masks.

The model performs well on dominant classes but struggles with smaller classes. This is mainly due to class imbalance in the dataset.

7. CONCLUSION

The project successfully demonstrates semantic segmentation using U-Net.

The combination of Dice Loss and CrossEntropy Loss improves segmentation performance.

The model achieved a peak validation accuracy of **74.78%** and a maximum IoU score of approximately **0.45**, demonstrating effective segmentation performance, especially for dominant terrain classes.

8. FUTURE WORK

- Larger datasets
- GPU training
- Advanced data augmentation techniques.